APPLIED CHEMISTRY (2023) 1. Chemical unit processes: CHEM-321 (SEMESTER-VI)

Elementary treatment of general unit Chemical unit processes, Nitration; Halogenation; Sulfonation; Hydration and Hydrogenation and Oxidation with real examples from Industry.

SHORT QUESTIONS:

Q.1: Define UNIT OPERATION with example.

Ans: A unit operation is a single physical step or change, such as separation, crystallization, evaporation, filtration, or heat transfer, which are integral parts of chemical manufacturing and processing.

- In chemical engineering and the chemical industries, a **unit operation** is a fundamental step in a process, involving a physical change or in some cases a chemical transformation.
- A unit operation is the building block of chemical processes.
- A set of experimental techniques associated with unit processes is called unit operations.
- Unit operation is the science of engineering aims at breaking up a complex process into a sequence of individual physical steps.

Examples of Unit Operations:

i) **Distillation**: Separation of components due to their boiling points difference. i.e. Separating ethanol from a water-ethanol mixture.

ii) **Filtration**: Removal of solid particles from a fluid by passing it through a porous medium. i.e. removing solids from wastewater.

iii) **Evaporation**: Removing a solvent to concentrate the solution. i. e. sugar cane juice is concentrated to produce crystals.

iv) **Crystallization**: Formation of crystals from a concentrated solution on cooling. i.e. Producing sugar crystals from concentrated Sugar-Cane Juice.

v) **Drying**: Removing moisture from a solid material. i.e. Drying wet pharmaceutical powders.

vi) **Absorption**: Transfer of a substance from one phase to another phase. i.e. Absorbing ammonia gas in water to form ammonium hydroxide.

Some other examples of unit operations are: **Adsorption, Heat Exchanging, Mixing, Extracting, etc.**

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Q.2: Define UNIT CHEMICAL PROCESSES with examples.

Ans: Unit chemical processes are fundamental chemical reactions that are used in chemical engineering and industrial chemistry to transform raw materials into desired products. Chemical process or unit process is the chemical transformation or conversion that is performed in the process. It produces entirely new substance(s) having properties quite different from those of the original substance(s). These processes involve specific types of chemical transformations, such as oxidation, reduction, polymerization, hydrolysis, etc.

- Each unit chemical process can be a part of larger manufacturing systems or used in various combinations to achieve complex chemical production goals.
- A combination of unit process and unit operation constitutes an industrial process.

Examples of some common unit chemical processes:

i) **Oxidation**: A chemical process in which a substance loses electrons, often involving the addition of oxygen or the removal of hydrogen. e.g. The oxidation of ethylene to produce ethylene oxide, which is a precursor for antifreeze and polyester production. $CH_2{=}CH_2 + \frac{1}{2}O_2 \xrightarrow{Ag\ (300\ ^oC)} CH_2{-}CH_2$ (with O bridging the two CH_2)

ii) **Reduction**: A chemical process in which a substance gains electrons, often involving the removal of oxygen or the addition of hydrogen. e.g. The reduction of iron ore (iron oxide) to produce metallic iron in a blast furnace. $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$

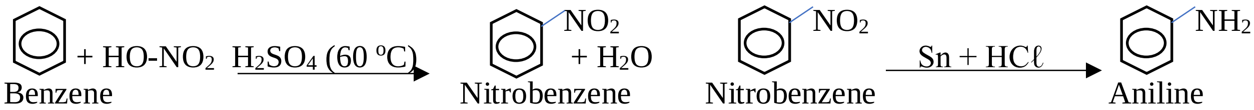
iii) **Hydrolysis**: A chemical reaction involving the breaking of a bond in a molecule using water. e.g. The hydrolysis of esters to produce alcohols and carboxylic acids, such as the hydrolysis of ethyl acetate to ethanol and acetic acid. $CH_3COOC_2H_5 + H_2O \xrightleftharpoons{NaOH} CH_3COOH + C_2H_5OH$

iv) **Polymerization**: A process in which small molecules (monomers) combine to form a large chain-like molecule (polymer). e.g. The polymerization of ethylene to produce polyethylene, a common plastic used in packaging. $nCH_2{=}CH_2 \rightarrow (\!-CH_2 - CH_2-\!)_n$

v) **Hydrogenation**: A chemical reaction between hydrogen and a compound in the presence of a catalyst. e.g. The hydrogenation of vegetable oils to produce margarine and vegetable ghee.

$$\begin{array}{l} CH_2\text{-}O\text{-}CO\text{-}(CH_2)_7\text{-}CH{=}CH\text{-}(CH_2)_7\text{-}CH_3 \\ | \\ CH\text{-}O\text{-}CO\text{-}(CH_2)_7\text{-}CH{=}CH\text{-}(CH_2)_7\text{-}CH_3 \\ | \\ CH_2\text{-}O\text{-}CO\text{-}(CH_2)_7\text{-}CH{=}CH\text{-}(CH_2)_7\text{-}CH_3 \end{array} + 3H_2 \xrightarrow{Ni} \begin{array}{l} CH_2\text{-}O\text{-}CO\text{-}(CH_2)_{16}\text{-}CH_3 \\ | \\ CH_2\text{-}O\text{-}CO\text{-}(CH_2)_{16}\text{-}CH_3 \\ | \\ CH_2\text{-}O\text{-}CO\text{-}(CH_2)_{16}\text{-}CH_3 \end{array}$$

vi) Nitration: A chemical process in which a nitro group ($-NO_2$) is introduced into an organic molecule. e.g. The nitration of benzene to produce nitrobenzene, used to make aniline for dyes and pharmaceuticals etc.



vii) Sulfonation: The introduction of a sulfonic acid group ($-SO_3H$) into an organic molecule. e.g. The sulfonation of benzene to produce benzene sulfonic acid, used in detergents and surfactants.

$$C_6H_6 + HOSO_3H \xrightarrow{80\ ^\circ C} C_6H_5SO_3H + H_2O$$

viii) Alkylation: A process in which an alkyl group is introduced into an organic molecule. e.g. The alkylation of isobutane with butene to produce high-octane gasoline components.

$$\underset{\text{Isobutane}}{(CH_3)_3CH} + \underset{\text{Isobutylene}}{CH_2 = C(CH_3)_2} \xrightarrow{H_2SO_4} \underset{\text{Isooctane (2,2,4-trimethylpentane)}}{(CH_3)_3C-CH_2-CH(CH_3)_2}$$

ix) Esterification: A reaction between an acid and an alcohol to form an ester and water. e.g. The esterification of acetic acid with ethanol to produce ethyl acetate, a solvent used in coatings and adhesives

$$CH_3COOH + C_2H_5OH \overset{H_2SO_4}{\rightleftharpoons} CH_3COOC_2H_5 + H_2O$$

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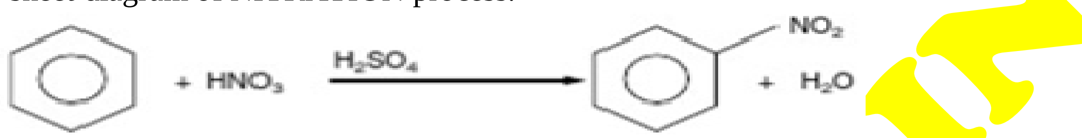
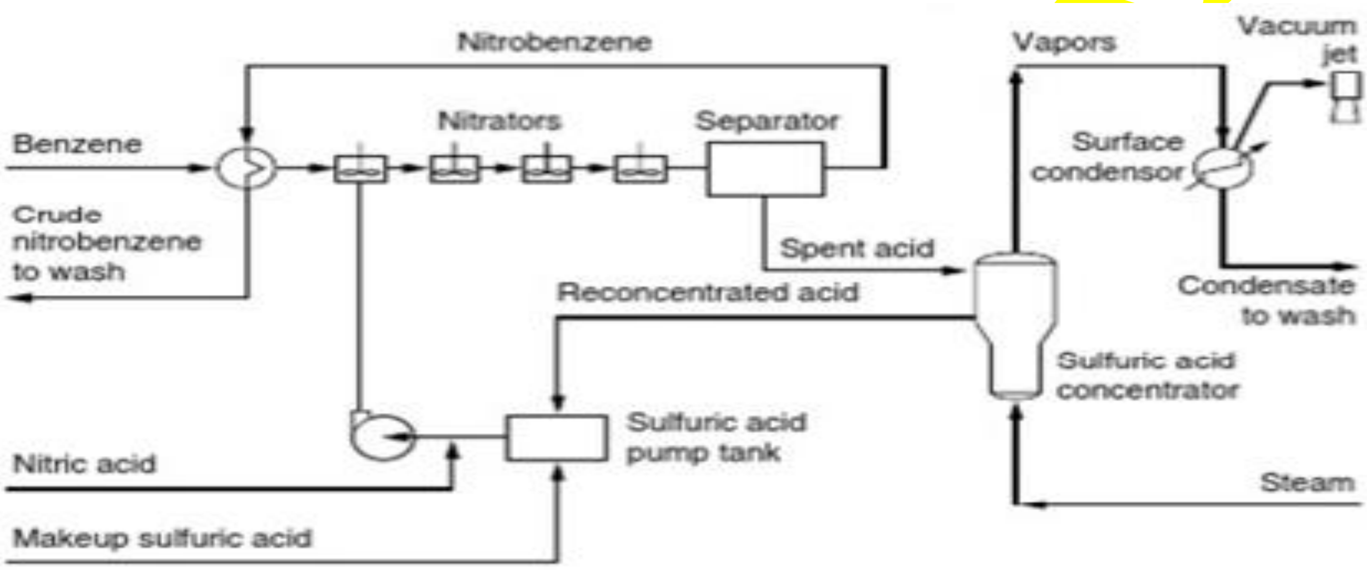
Q.3: Draw the flow sheet diagram of NITRATION process.





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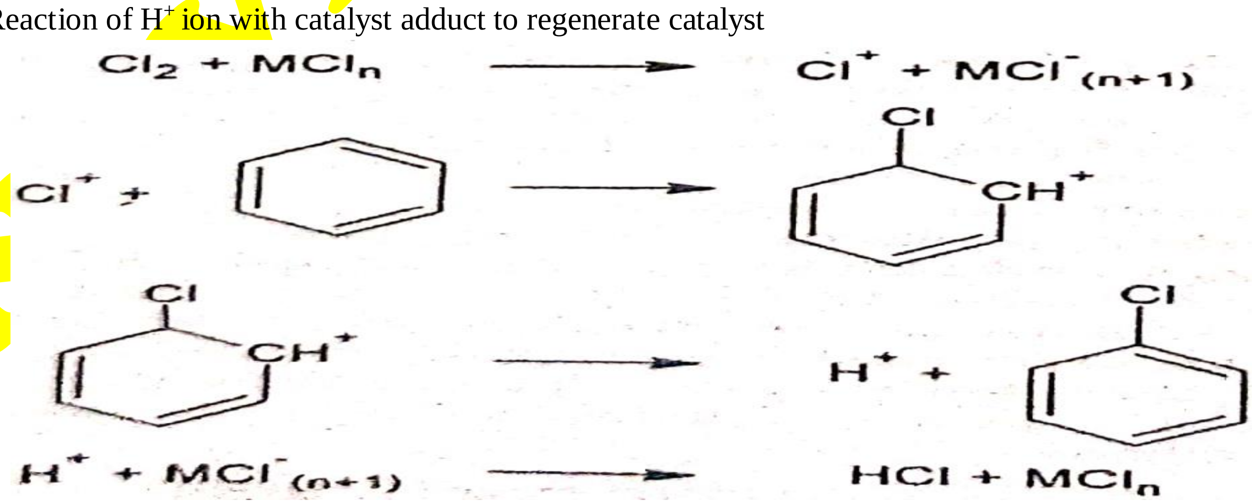
Q.4: Give the mechanism of CHLORINATION PROCESS of benzene.

Ans: Electrophilic aromatic substitution is performed with the aid of a Friedel-crafts catalyst (halide salt of metal) to give halo-aromatic compounds. Metal halides are electron deficient and act as a Lewis acid. Mechanism consists of following steps.

i) Generation of electrophile (chloronium ion)
ii) Attack of electrophile (Chloronium ion) on benzene ring to form benzenonium ion
iii) Replacement of H^+ ion to substitution product
iv) Reaction of H^+ ion with catalyst adduct to regenerate catalyst



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Q.5: Describe any two SULFONATING agents.

Ans: Some important sulfonating agents:

i) Sulfur trioxide (SO_3): Sulfur trioxide is directly added to the compound. It is principally used for sulfonating aromatic compounds (e.g. toluene, nitrobenzene, detergent alkylate), petroleum oils and long chain terminal alkanes.

$$RH + SO_3 \rightarrow RSO_2OH$$

ii) Sulfuric acid (H_2SO_4) and Oleum ($H_2SO_4.SO_3$): Sulfuric acid is the hydrate of sulfur trioxide ($H_2O.SO_3$) and the most common reagent for direct sulfonation. Oleum is a solution of sulfur trioxide in 100% sulfuric acid.

$$RH + H_2SO_4 \rightleftharpoons RSO_2OH + H_2O$$

iii) Chlorosulfonic acid ($ClSO_3H$): It is an adduct of sulfur trioxide and hydrochloric acid is released as a gas during sulfonation. Excess of chlorosulfunic acid is used for sulfonyl chlorides. It is used for sulfating alcohols and to prepare sulfonyl chlorides.

$$RH + ClSO_3H \rightarrow RSO_2OH + HCl$$
$$RSO_2OH + ClSO_3H \rightleftharpoons RSO_2Cl + H_2SO_4$$

iv) Amido sulfuric acid (sulfamic acid, NH_2SO_2OH): Amidosulfuric acid is used for sulfation and sulfamation reactions.

$$ROH + NH_2SO_2OH \rightarrow ROSO_2ONH_4$$
$$2RNH_2 + NH_2SO_2OH \rightarrow RNHSO_2OH + NH_3$$

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Q.6: Why agitation is important during SULPHONATION process?

Ans: Sulfonation reaction mixtures are comparatively viscous and cannot heat exchange sufficiently in required time leading to poor product quality or reduced productivity. To overcome this problem, sulfonators are equipped with two common types of agitators: **i)** propeller **ii)** turbine

These two are used for reaction mixtures of low or moderate viscosity.

For more viscous products, a close fitting anchor agitator equipped with supplemental vertical arms especially blades having knife-edges on the leading sides.

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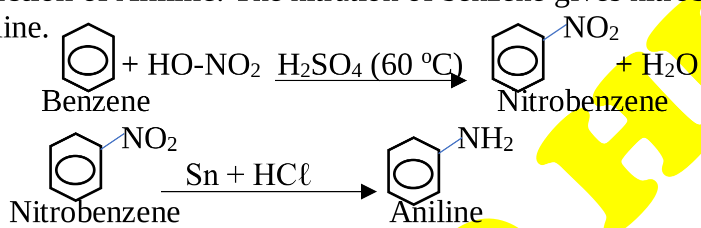
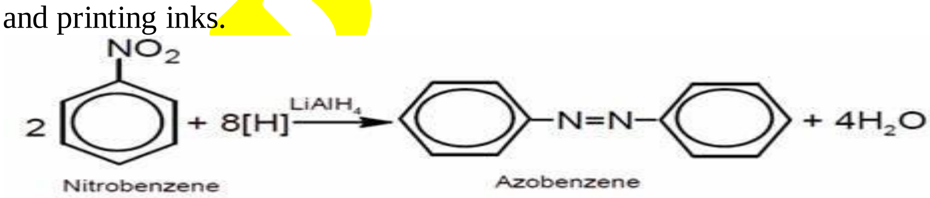
Q.7: Mention the major application of NITRATION of benzene.

Ans: Major applications of nitration of benzene are:

i) **Production of Aniline**: The nitration of benzene gives nitrobenzene, which is an intermediate in the synthesis of aniline.

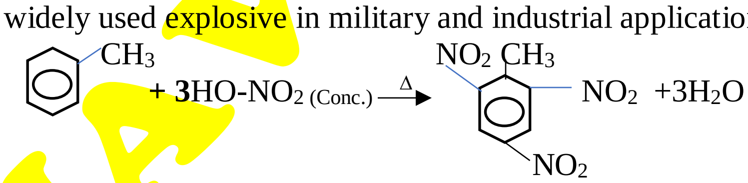


a. **Dye Industry**: Aniline is a primary raw material for the production of aniline dyes and pigments, used in textiles, leather, and printing inks.

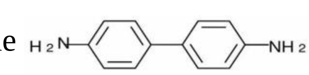


b. **Pharmaceuticals**: Aniline derivatives are used in the synthesis of various pharmaceuticals, including paracetamol (acetaminophen), sulfa drugs, and several other medicinal compounds.

ii) **Production of Explosives, TNT (Trinitrotoluene)**: Nitration of toluene produces trinitrotoluene (TNT), a widely used explosive in military and industrial applications.



iii) **Solvents and Chemical Intermediates**: Nitrobenzene itself is used as a solvent and as an intermediate in the synthesis of lubricating oils, dyes, and synthetic resins.

iv) **Perfumes and Flavorings**: Nitrobenzene is also used in the synthesis of benzidine  used in the manufacture of dyes (now obsolete due to toxicity)

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Q.8: Give two industrial applications of HYDROGENATION?

Ans: '**Hydrogenation** is a chemical reaction between molecular hydrogen (H_2) and another compound or element, usually in the presence of a catalyst such as nickel (Ni), Palladium (Pd), or Platinum (Pt).' So, it is also called catalytic hydrogenation.

- The process is commonly employed to reduce unsaturated organic compounds.
- Hydrogenation is an exothermic process.

Applications of hydrogenation:

Hydrogenation is widely used in the industry. The synthesis of **Ammonia, fuels (hydrocarbons), alcohols, pharmaceuticals, margarine, polyols, various polymers and chemicals** (hydrogen chloride and hydrogen peroxide) are products treated using a hydrogenation process.

i) Hydrogenation of Fats and Oils to produce Vegetable ghee:

Hydrogenation is used to convert unsaturated fats and oils (liquids at room temperature) into saturated fats (solids at room temperature). This process produce margarine, shortening, and vegetable ghee from vegetable oils.

ii) **Petrochemical Industry**: In petroleum refining, hydrogenation is used to remove impurities such as sulfur, nitrogen, and oxygen from hydrocarbons, a process known as hydrotreating.

$$3H_2 + N_2 \rightleftharpoons 2NH_3 \qquad H_2 + S \rightarrow H_2S$$

Hydrocracking involves breaking down large, heavy molecules into lighter, more valuable products like gasoline, kerosene, and diesel.

$$\underset{\text{n-Decane}}{CH_3(CH_2)_8CH_3} + H_2 \xrightarrow{\text{Catalyst}} \underset{\text{n-Pentane}}{2CH_3(CH_2)_3CH_3}$$

iii) **Chemical Industry**: **Production of Ammonia (Haber Process)**: The hydrogenation process is used to synthesizes ammonia from nitrogen and hydrogen. Ammonia is a key component in fertilizers and various chemicals.

$$3H_2 + N_2 \underset{450\,^{o}C,\ 200\ atm}{\overset{Fe\,(Al_2O_3 + MgO)}{\rightleftharpoons}} 2NH_3$$

Hydrogenation of Nitriles to Amines: Hydrogenation is used to convert nitriles to primary amines, which are important intermediates in the production of polymers, pharmaceuticals, and agrochemicals.

$$\underset{\text{Nitrile}}{R - C \equiv N} \xrightarrow{LiAlH_4} \underset{\text{Primary amine}}{R - CH_2 - NH_2}$$

iv) **Metallurgical Applications**: In metallurgy, hydrogenation can be used to reduce metal ores to pure metals. This is an alternative to traditional carbon-based reduction methods and can produce metals with fewer impurities.

Metal oxide + Hydrogen → Metal + Steam

v) **SUBSTITUTE NATURAL GAS (SNG):** The low- and medium-heat syngas produced from coal can be converted to a high-heat content gas (30 to 37 MJ/m³) similar to natural gas by the following reactions:

$C + H_2O \rightarrow CO + H_2$	gasification	(1)
$CO + H_2O \rightarrow CO_2 + H_2$	water gas shift reaction	(2)
$C + CO_2 \rightarrow 2CO$	Boudouard reaction	(3)

At sufficiently high pressures the hydrogen from reactions (1) and (2) will hydrogenate some of the carbon to yield methane.

$C + 2H_2 \rightarrow CH_4$		(4)
$CO + 3H_2 \xrightarrow{\text{catalyst}} CH_4 + H_2O$	methanation	(5)

The gas thus produced was known originally as synthetic natural gas, but now called substitute natural gas.

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Q.9: Give two industrial applications of HYDRATION.

Ans: **HYDRATION:-** A chemical reaction in which a substance chemically combines with water is called hydration reaction. In organic chemistry, the addition of water to unsaturated substances (alkene or alkyne) is called hydration reaction. Addition of water to a compound or substance is commonly used in various industrial applications to achieve specific objectives. Here are two significant industrial applications of hydration:

Cement Production: Hydration plays a fundamental role in the setting and hardening of cement. When water is added to cement powder, a series of exothermic chemical reactions occur, leading to the hydration of cement compounds such as tricalcium silicate (C3S), dicalcium silicate (C2S), tricalcium aluminate (C3A), and tetracalcium aluminoferrite (C4AF). The hydration process forms calcium silicate hydrates (C-S-H) and calcium hydroxide (CH) gels, which contribute to the binding and hardening of the cement paste.

Food and Beverage Industry:

Baking: Hydration is a key step in the formation of dough from flour in baking. When water is added to flour, proteins in the flour (such as gluten) hydrate and form a network that gives dough its elasticity, strength and texture of bread and other baked goods.

Production of Hydrated Lime: One significant application of hydration is in the production of hydrated lime (calcium hydroxide, $Ca(OH)_2$) from quicklime (calcium oxide, CaO). This process is commonly used in agriculture, and water treatment. $CaO + H_2O \rightarrow Ca(OH)_2$

Production of Hydrated Alumina: The hydration is utilized in the production of hydrated alumina (aluminum hydroxide, $Al(OH)_3$) from alumina (aluminum oxide, Al_2O_3) in the aluminum industry.

$$Al_2O_3 + 3H_2O \rightarrow 2Al(OH)_3$$

Hydrated alumina is processed to obtain aluminum metal by Bayer process and Hall-Héroult process.

Hydration in organic chemistry:

Ethylene Glycol:

Reaction of ethylene oxide with water yields ethylene glycol.

$$\underset{\diagdown\ O\ \diagup}{CH_2 - CH_2} + H_2O \rightarrow HO - CH_2 - CH_2 - OH$$

A large excess of water is supplied to the reactor to favour ethylene glycol formation over that of diethylene glycol and triethylene glycol, as the glycols are more reactive with ethylene oxide than is water.

Ethylene glycol is used as:

i) antifreeze (Almost 43 percent of glycol is consumed in this category)

ii) polyester fibre formation (Almost 26 percent of glycol is consumed in this process)

Ethanol:

Direct hydration of ethylene produces ethanol industrially.

$$CH_2 = CH_2 + H_2O \xrightarrow{H_3PO_4} CH_3 - CH_2 - OH$$

Ethylene and high temperature steam are mixed and passed over an acidic catalyst, usually phosphoric acid on a support at 250 to 300°C under pressure.

Synthetic ethanol is used:

i) as a chemical intermediate (for ethyl acetate, ethyl acrylate, glycol ethers, ethyl amines, etc.)

ii) in toiletries and cosmetics **iii)** as a solvent **iv)** as a raw material for vinegar **v)** in household cleaners, in detergents, in pharmaceuticals, in printing inks etc.

ACETALDEHYDE:

Acetaldehyde was originally made by the hydration of acetylene over an oxidation–reduction catalyst, mercurous–mercuric sulphate buffered by ferric sulphate. Vinyl alcohol is assumed to form momentarily and rearrange to acetaldehyde at atmospheric pressure and 95°C.

$$\underset{\text{Acetylene}}{HC \equiv CH + H_2O} \xrightarrow[\text{95 oC}]{\text{HgSO4 (1\%)/H2SO4}} \underset{\text{Vinyl alcohol}}{CH_2 = CHOH} \overset{\text{rearrangement}}{\rightleftarrows} \underset{\text{Acetaldehyde}}{CH_3CHO}$$

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Q.10: What is the state of BUTADIENE at atmospheric conditions?

Ans: Butadiene (1,3-butadiene) [CH_2=CH-CH=CH_2] is a colorless gas. It has a faint aromatic odor and is highly flammable under normal atmospheric conditions (1 atm pressure and around 25°C or 77°F). The properties of 1,3-butadiene are:

IUPAC name:	buta-1,3-diene	Density:	0.614 gcm^{-3}
Molar mass:	54 $gmol^{-1}$	Solubility in Water:	1.3 gdm^{-3}
Physical State:	Colorless gas	Odor:	aromatic
Boiling Point:	-4.4°C (24°F)	Melting Point:	-108.9°C (-164°F)
Critical Temperature:	152°C (306°F)	Critical Pressure:	4.25 MPa (618 psi)

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